

Rank, Nullity, and Row Reduction

Math 5250 — Week 1

Matrices

An $n \times m$ matrix over $\mathbf{R}$ is an array with n rows and m columns; the set of these is $M_{n \times m}(\mathbf{R})$ (similarly $M_{n \times m}(\mathbf{C})$ for complex arrays).

- ▶ $M_{n \times m}(\mathbf{R})$ is a vector space of dimension nm , with addition and scalar multiplication entry by entry: $(A + B)_{ij} = a_{ij} + b_{ij}$ and $(\lambda A)_{ij} = \lambda a_{ij}$.
- ▶ Elements of $\mathbf{R}^n$ may be viewed as $n \times 1$ column vectors or $1 \times n$ row vectors; by convention we use $n \times 1$ column vectors.

Matrix Multiplication

For A an $n \times m$ matrix and B an $m \times k$ matrix, the product $C = AB$ is $n \times k$ with entries

$$c_{ij} = \sum_{r=1}^m a_{ir} b_{rj}.$$

- ▶ Matrix multiplication is associative: $(AB)C = A(BC)$.
- ▶ It distributes over addition: $(A + B)C = AC + BC$ and $A(B + C) = AB + AC$.
- ▶ It is **not** commutative, even when both AB and BA are defined.

Linear Maps

Definition. $L : \mathbf{R}^m \rightarrow \mathbf{R}^n$ is **linear** if $L(x + y) = L(x) + L(y)$ and $L(ax) = aL(x)$.

- ▶ For an $n \times m$ matrix A , $L_A(x) = Ax$ defines a linear map $\mathbf{R}^m \rightarrow \mathbf{R}^n$.
- ▶ If A is $n \times m$ and B is $k \times n$, then $L_B \circ L_A = L_{BA}$.
- ▶ Everything above holds with $\mathbf{R}$ replaced by $\mathbf{C}$.

Key Subspaces and Invariants of a Matrix

Definition. The **column space** of $A \in M_{n \times m}(\mathbf{R})$ is the subspace of $\mathbf{R}^n$ spanned by the columns of A ; it equals the image of L_A . The **column rank** is its dimension.

Definition. The **nullspace** (or **kernel**) of A is $\{x \in \mathbf{R}^m : Ax = 0\}$, a subspace of $\mathbf{R}^m$; the **nullity** is its dimension.

Definition. The **row space** of A is the subspace of $\mathbf{R}^m$ spanned by the rows of A ; the **row rank** is its dimension.

Elementary Row Operations and Elementary Matrices

Three **elementary row operations**:

- ▶ Swap two rows.
- ▶ Multiply a row by a nonzero scalar.
- ▶ Replace $A_j.$ by $A_j. - \lambda A_i.$

Each corresponds to an **elementary matrix** E , with $A' = EA$; elementary matrices are invertible.

Proposition. If A' is obtained from A by elementary row operations, then A' and A have the same row space, the same nullspace, and the same column rank.

Row Reduced Echelon Form

Definition. A is in **row reduced echelon form** if:

- a. All-zero rows are at the bottom.
- b. The first nonzero entry (“pivot”) of each nonzero row is 1.
- c. Pivots move down and to the right.
- d. A pivot is the only nonzero entry in its column.

Proposition. Every matrix A can be transformed into row reduced echelon form by a sequence of row operations.

Rank and Nullity from RREF

Suppose A is $n \times m$, in row reduced echelon form, with r pivots.

Proposition.

- ▶ The row rank and column rank both equal r .
- ▶ The nullity equals $m - r$: for each non-pivot column j , a vector $x^{(j)} \in \ker A$ is obtained by putting 1 in position j , 0 in the other non-pivot positions, and $-A_{i,j}$ in each pivot position p_i . These $m - r$ vectors form a basis of $\ker A$.

The Rank–Nullity Corollary

Corollary. For any matrix A with m columns:

- ▶ The row rank and column rank of A agree; this common value is the **rank** of A .
- ▶ $\text{rank}(A) + \text{nullity}(A) = m$.

Algorithm: Testing Membership in the Range

To decide whether v is in the range of A :

- ▶ Form the augmented matrix $A' = [A, v]$.
- ▶ Row reduce A' to echelon form.
- ▶ If the last column contains a pivot, v is **not** in the range; otherwise it **is**.